



microring modulators.



contents

Ring Resonator Simulation

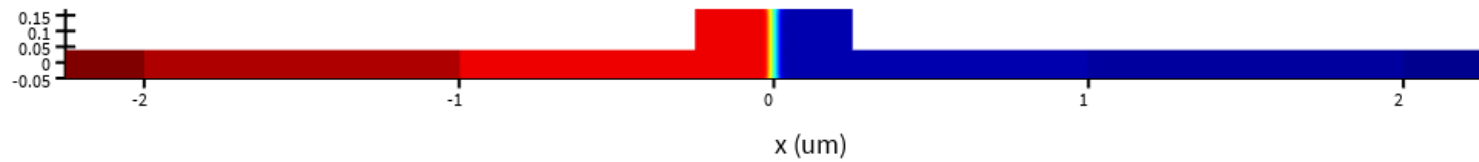
- Lateral PN junction.
- Vertical PN junction.



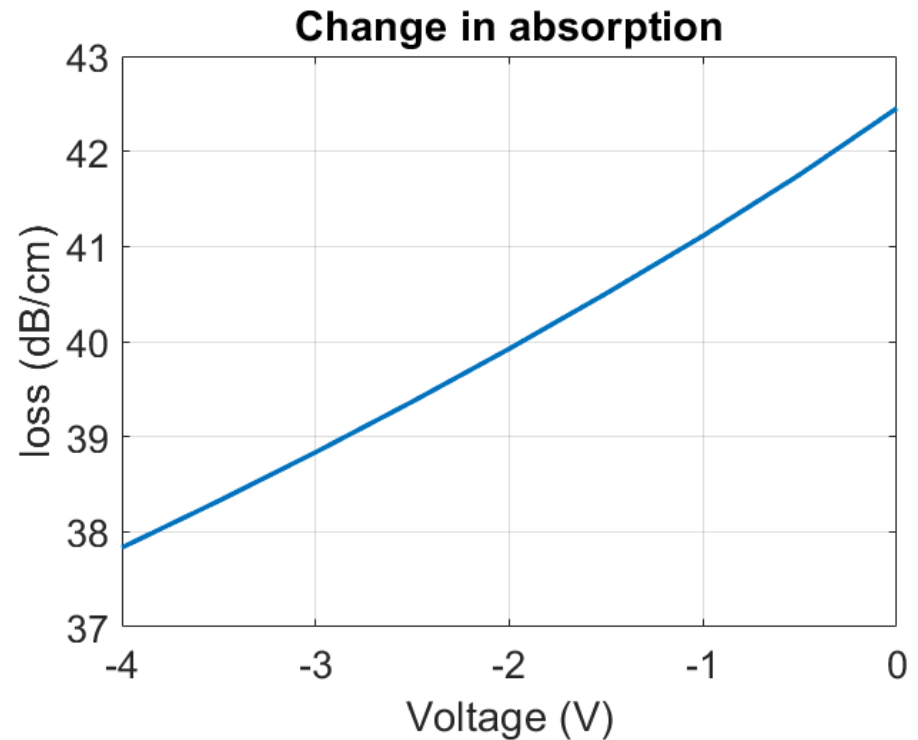
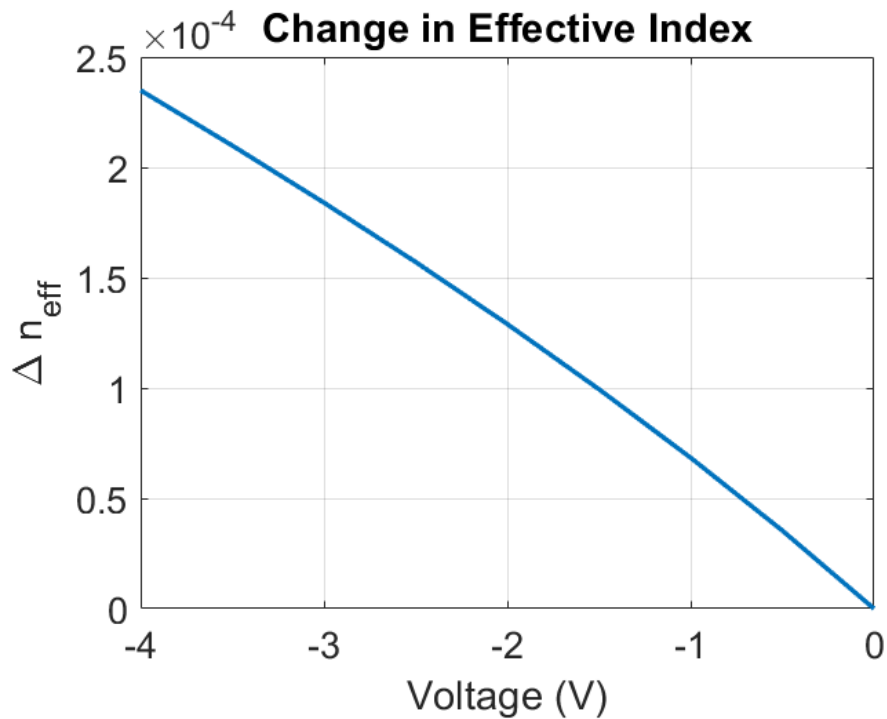
contents

- ➡ Lateral PN junction.

Lateral PN junction Simulation

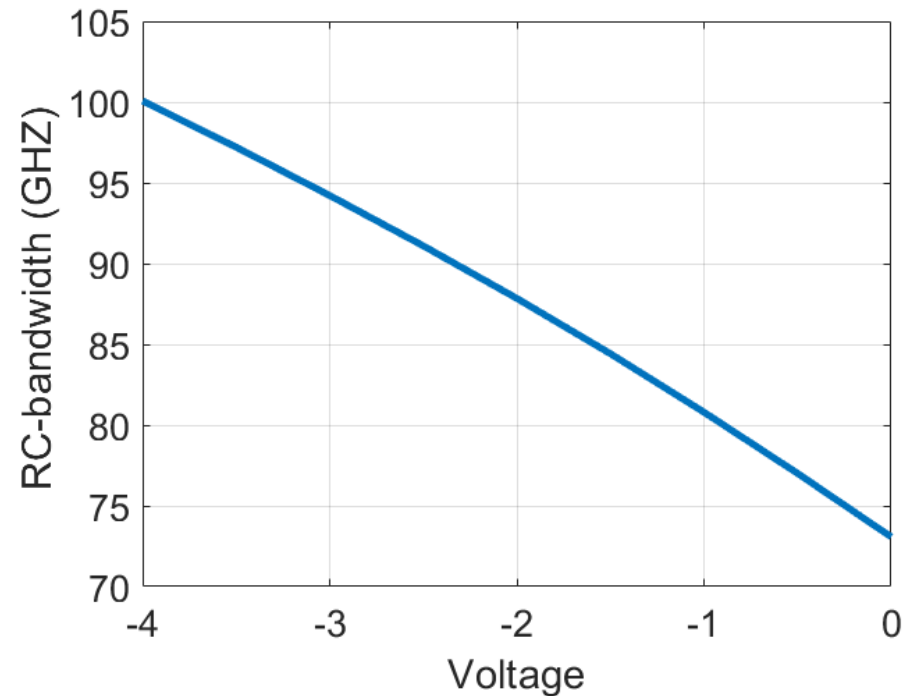
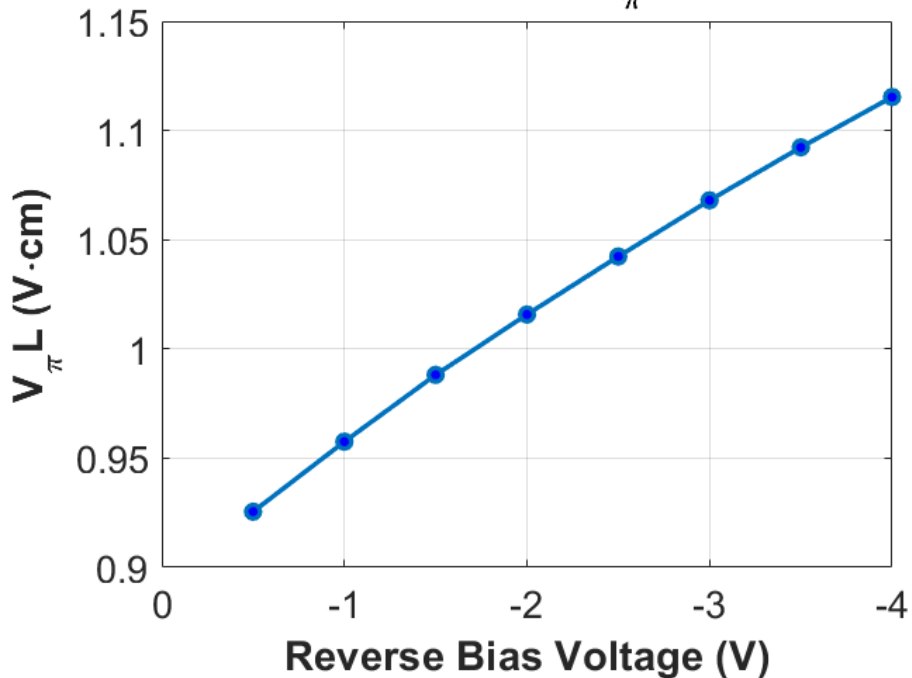


Lateral PN junction Simulation

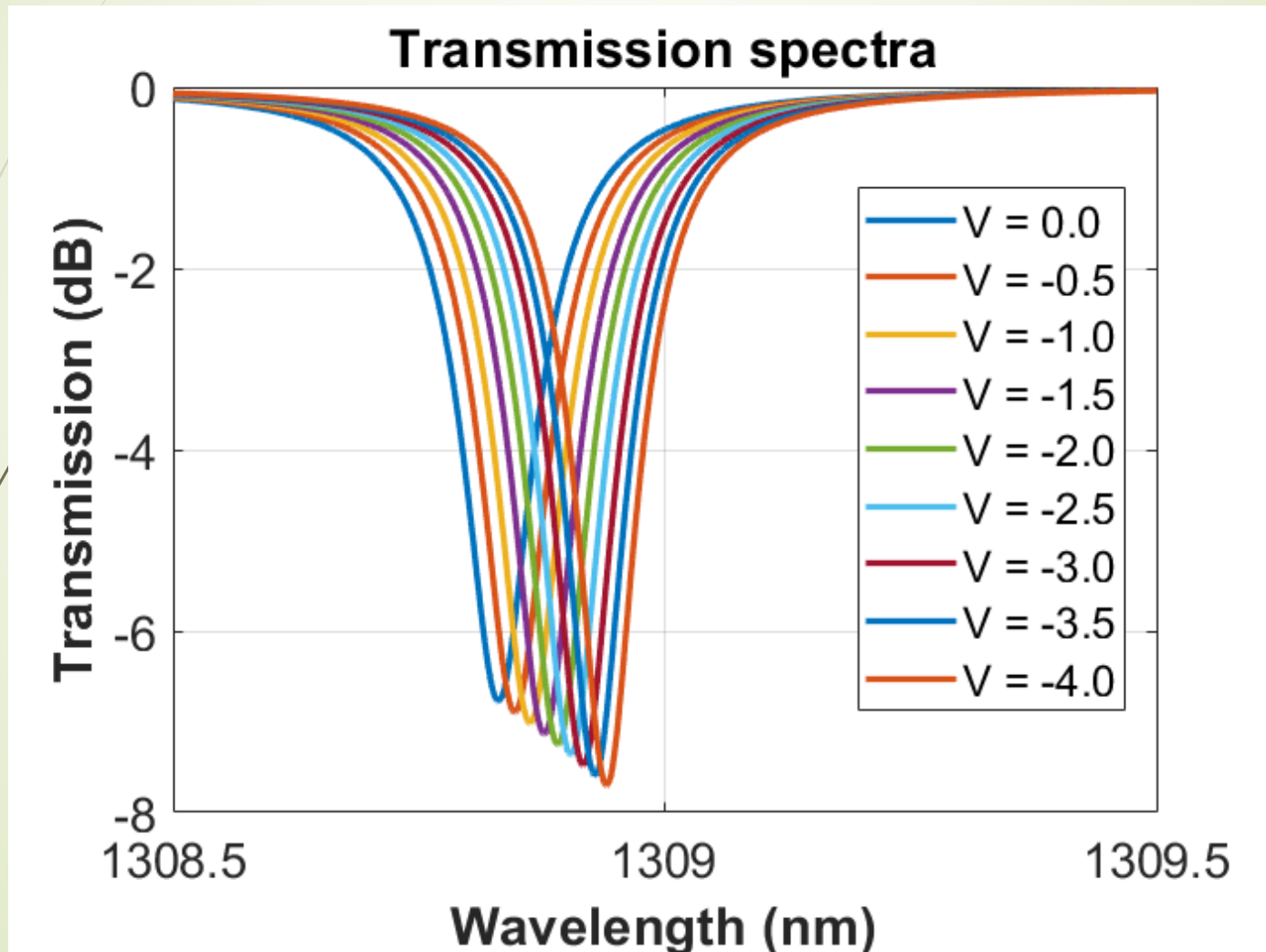


Lateral PN junction Simulation

Modulation Efficiency ($V_{\pi} L$) vs. Voltage



➤ Lateral PN junction used for modulation



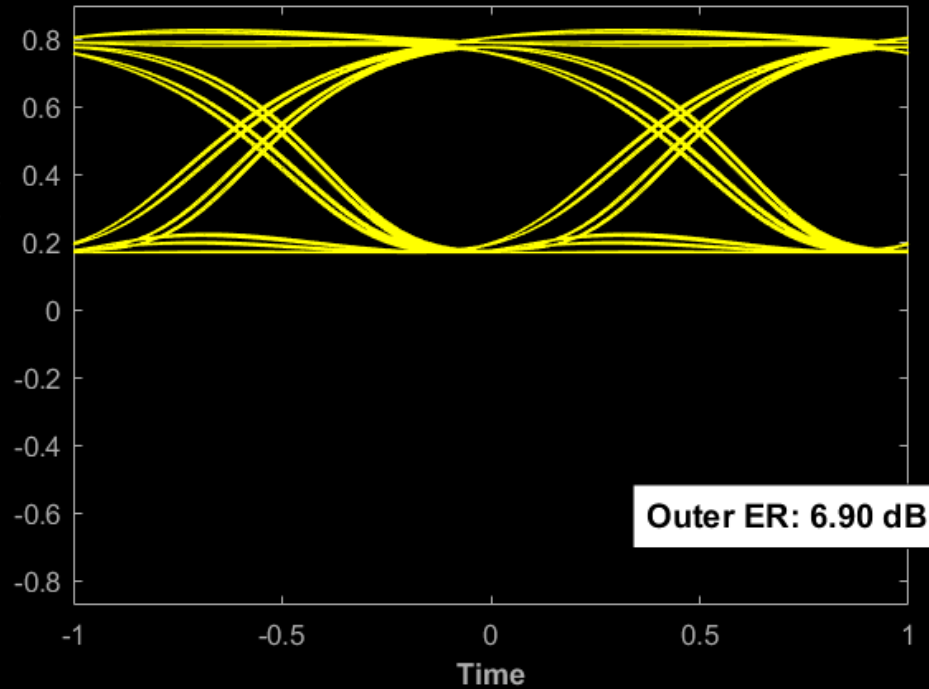


➤ Lateral PN junction used for modulation

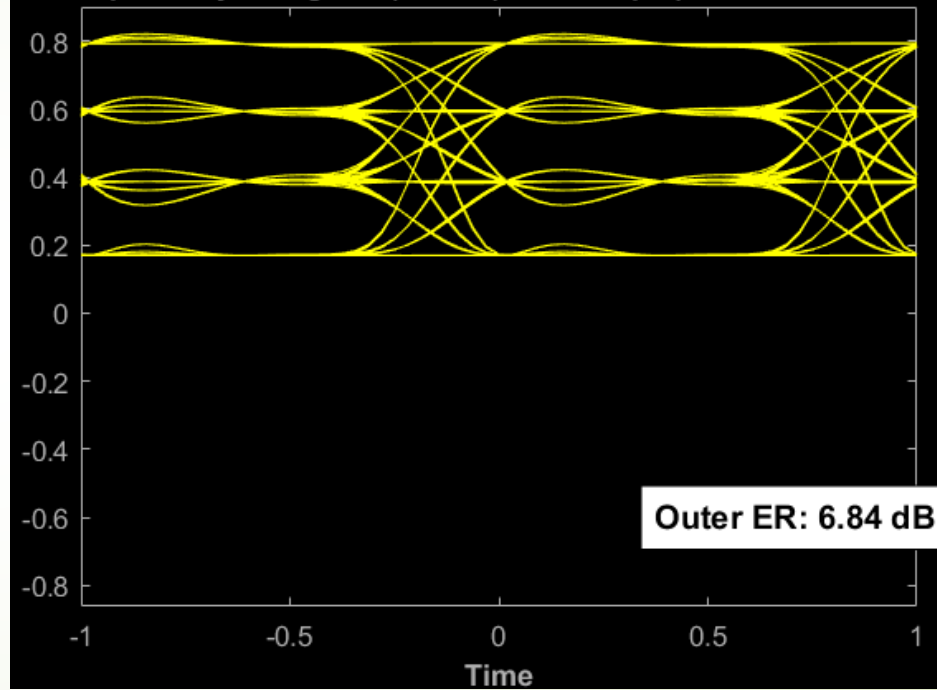
- **FSR = 6.98 nm,**
- **FWHM = 0.086 nm,**
- **3dB BW = 15.0467 GHz,**
- **Q = 15221.6,**
- **efficiency = 27.50 pm/V**

➤ Lateral PN junction used for modulation

Optical Eye Diagram (NRZ) at $\lambda = 1308.94$ nm



Optical Eye Diagram (PAM-4) at 50 Gbps | $\lambda = 1308.94$ nm





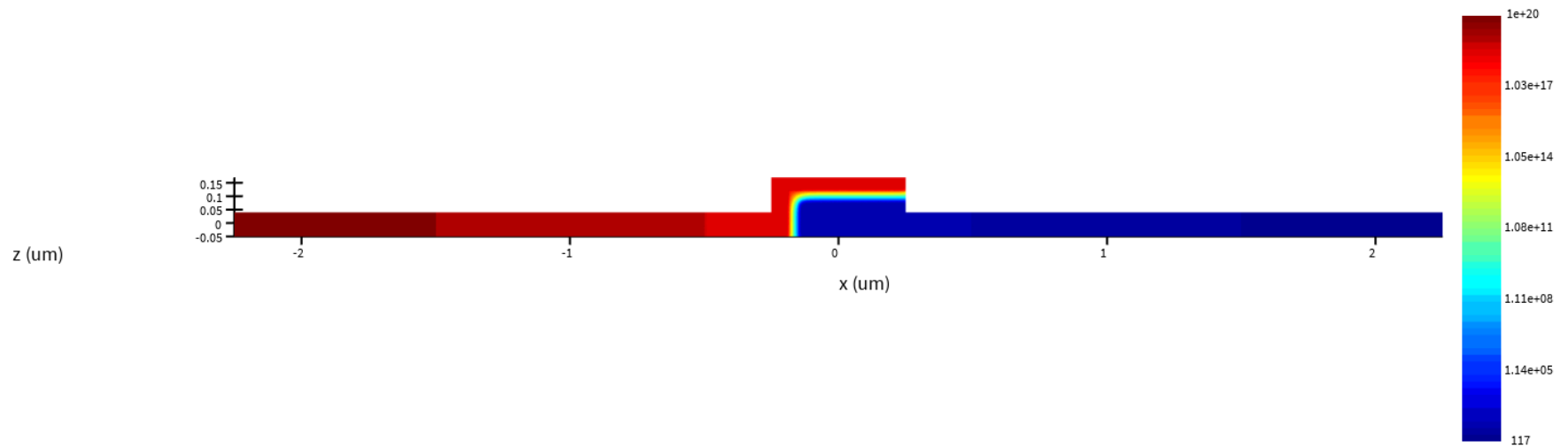
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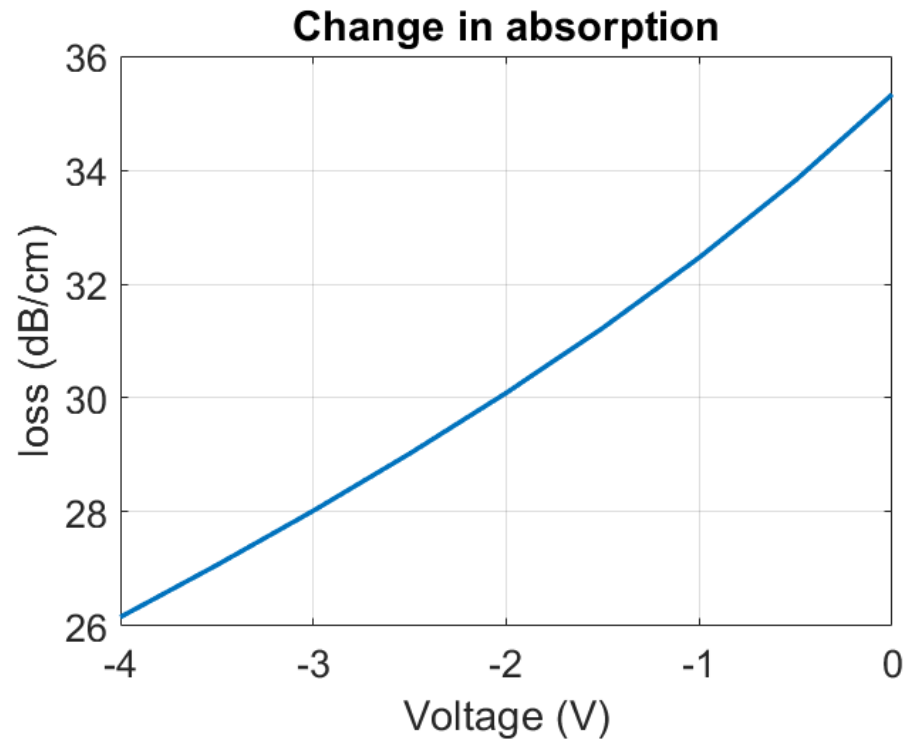
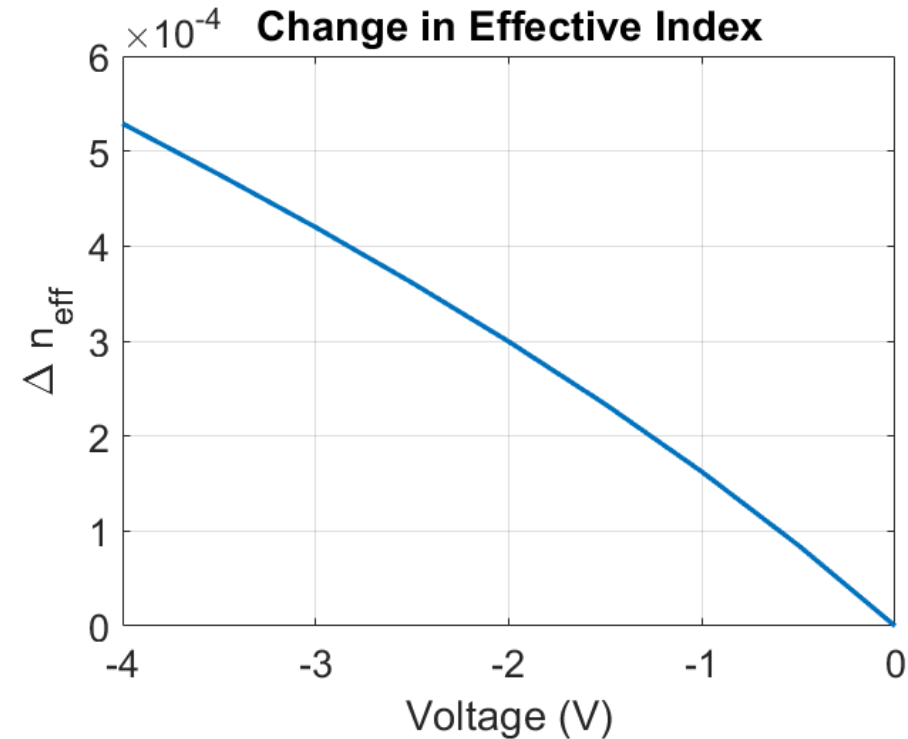
Vertical PN junction.



Vertical PN junction simulation

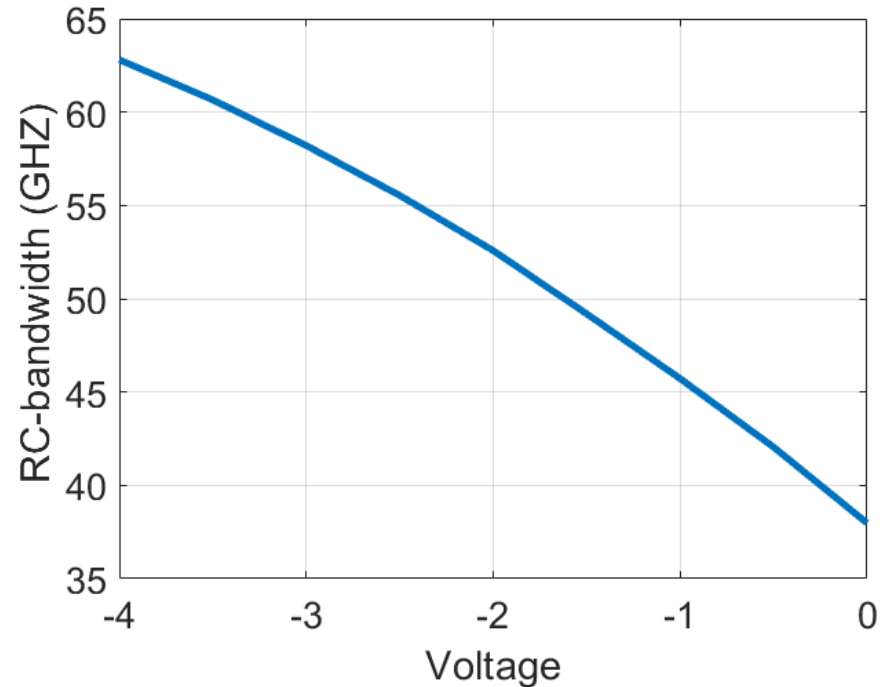
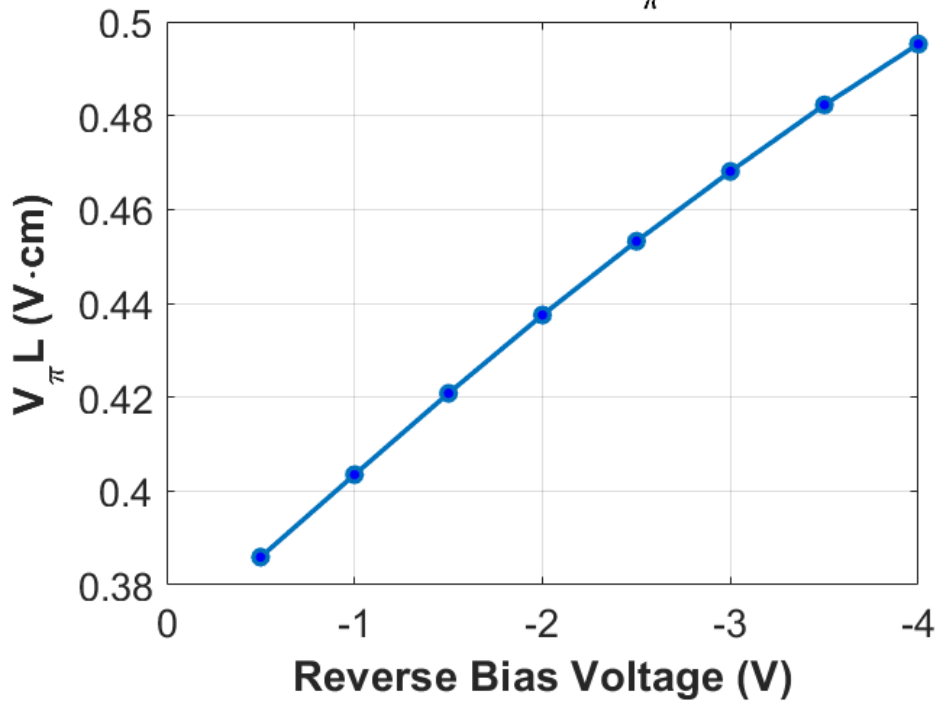


Vertical PN junction simulation

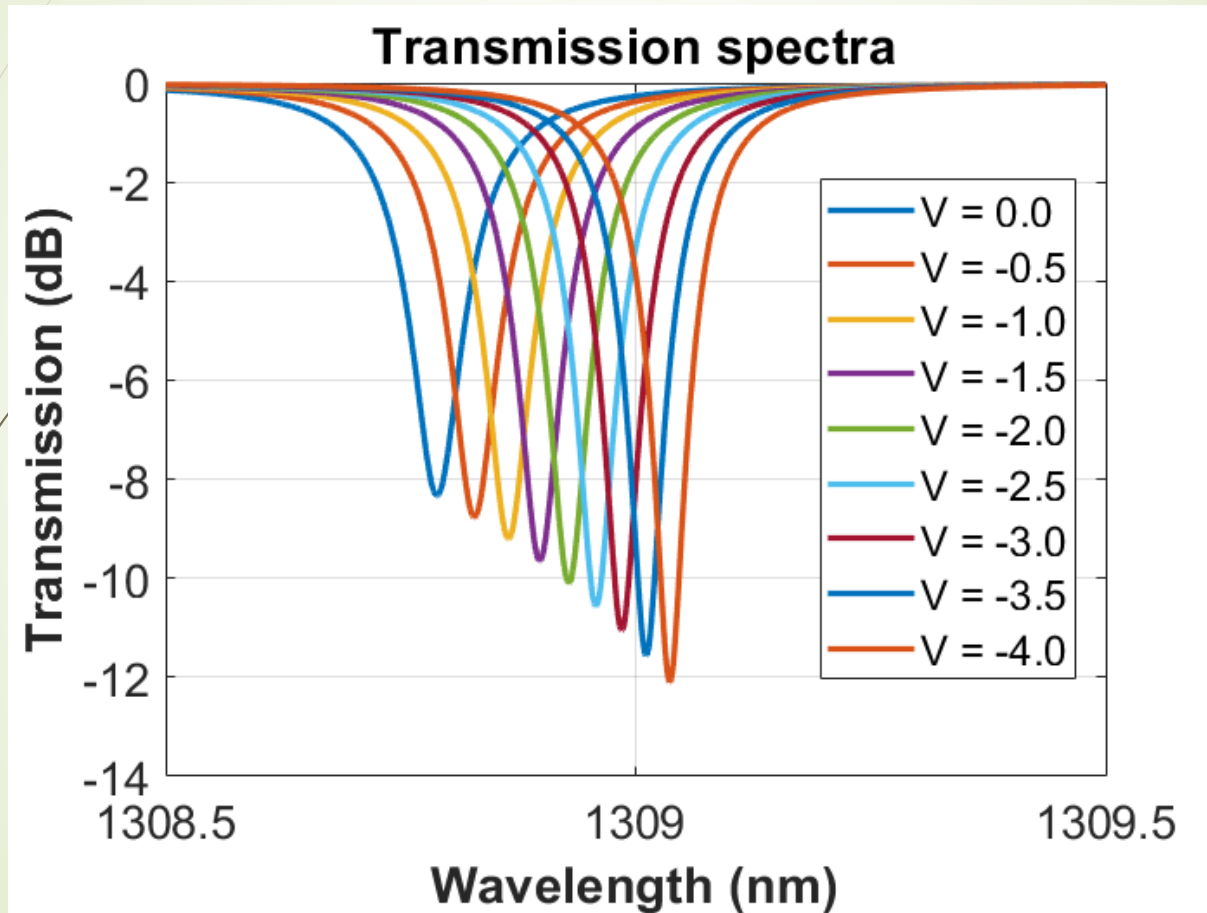


Vertical PN junction simulation

Modulation Efficiency ($V_{\pi} L$) vs. Voltage



- Vertical PN junction used for modulation

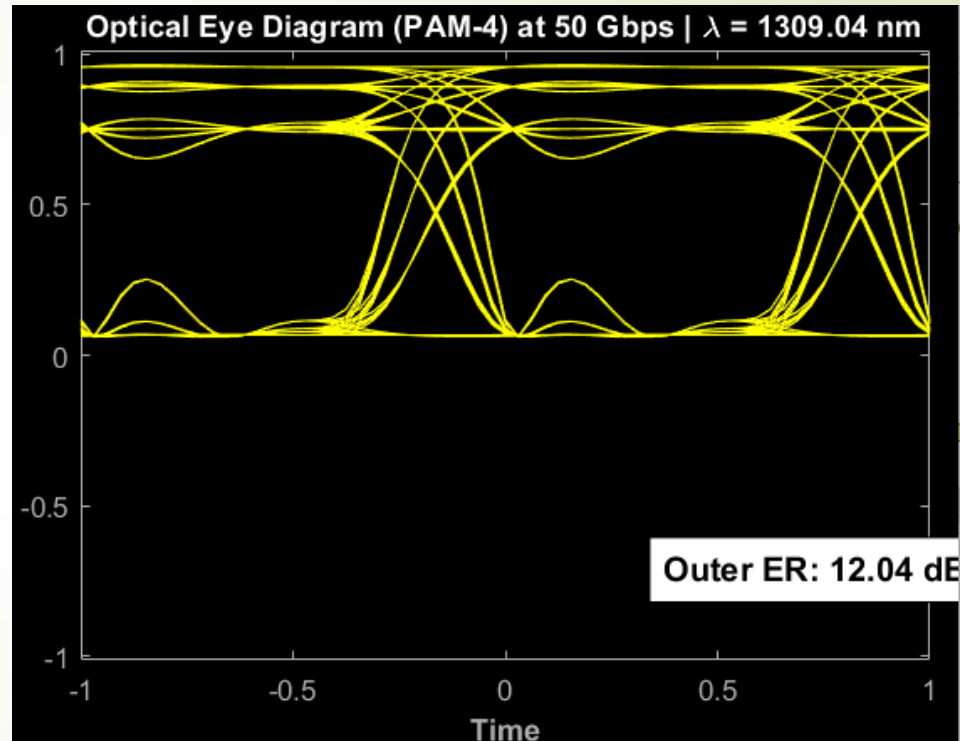
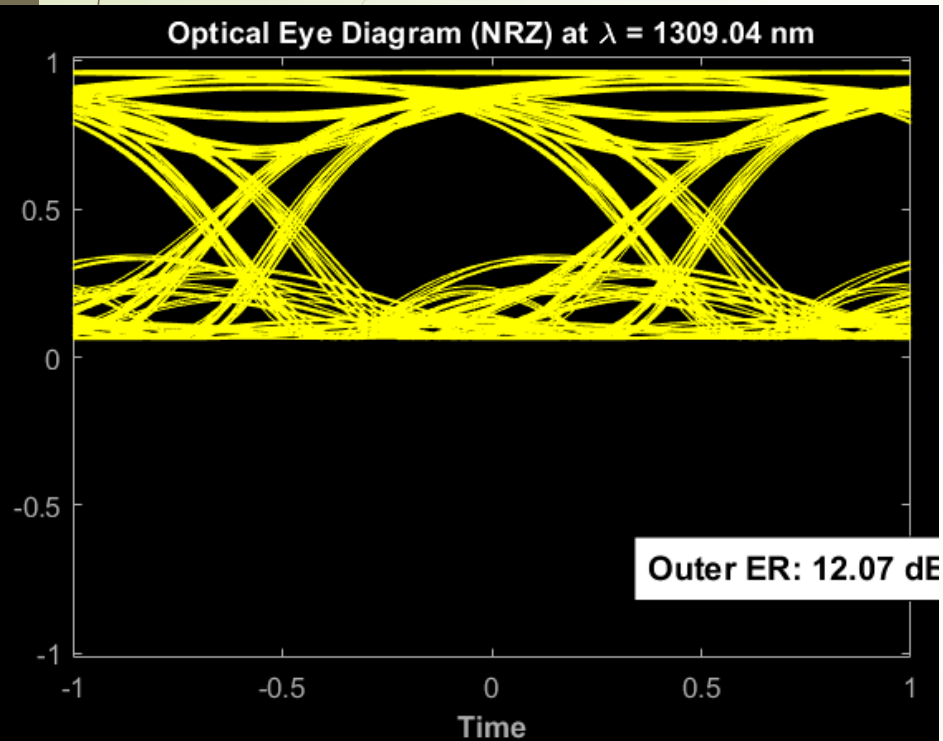


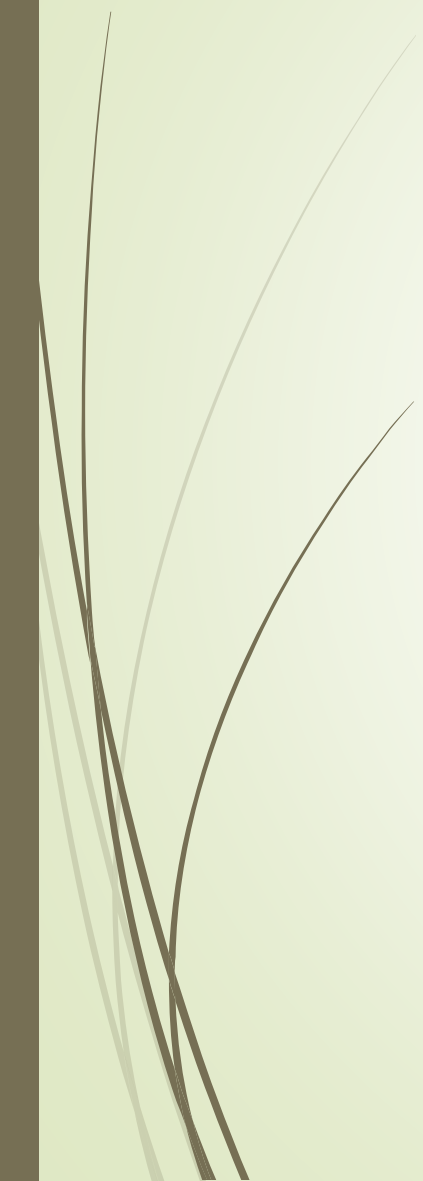


➤ Vertical PN junction used for modulation

- **FSR = 6.98 nm,**
- **FWHM = 0.067 nm,**
- **3dB BW = 11.7689 GHz,**
- **Q = 19459.6,**
- **efficiency = 61.75 pm/V**

Vertical PN junction used for modulation





Thanks 😊